INTRODUCTION TO COMPLEX NUMBERS – SHEET 1

Solutions will be made available online on Monday of Week 2. Students should not necessarily expect a college tutorial on these problems, and in this case students are asked to review and mark their own work.

1. Which of the following quadratic equations require the use of complex numbers to solve them?

$$3x^2 + 2x - 1 = 0; \qquad 2x^2 - 6x + 9 = 0; \qquad -4x^2 + 7x - 9 = 0.$$

2. Show directly that if $zw = 0$, where z, w are two complex numbers, then $z = 0$ or $w = 0$ (or both). Now re-prove this making use of properties of the modulus function.

3. Put each of the following numbers into the form $a + bi$.

$$(3 + 2i) + (2 - 7i), \qquad (1 + 2i)(3 - i), \qquad (1 + 2i)/(3 - i), \qquad (1 + i)^4.$$

4. Find the modulus and argument of each of the following numbers.

$$1 + \sqrt{3}i, \qquad (2 + i)(3 - i), \qquad (1 + i)^5, \qquad (1 + 2i)^3/(2 - i)^3.$$

5. Find the two square roots of $-5 - 12i$. Hence solve the quadratic equation $z^2 - (4 + i)z + (5 + 5i) = 0$.

6. On separate Argand diagrams sketch the following sets.

(i) $\lvert z\rvert < 1$;	(ii) $\operatorname{Re} z = 3$;	(iii) $\lvert z - 1\rvert = \lvert z + i\rvert$;
(iv) $\arg(z - i) = \pi/4$;	(v) $-\pi/4 < \arg z < \pi/4$;	(vi) $\operatorname{Re}(z + 1) = \lvert z - 1\rvert$;
(vii) $\lvert z - 3 - 4i\rvert = 5$;	(viii) $\operatorname{Re}((1 + i)z) = 1$;	(ix) $\operatorname{Im}(z^3) > 0$.

7. Let $z = \operatorname{cis}\theta = e^{i\theta}$ and let n be an integer.
(i) Show that $2\cos\theta = z + z^{-1}$ and that $2i\sin\theta = z - z^{-1}$.
(ii) Using De Moivre's theorem, show further that $2\cos n\theta = z^n + z^{-n}$ and that $2i\sin n\theta = z^n - z^{-n}$.
(iii) Deduce that $16\cos^5\theta = \cos 5\theta + 5\cos 3\theta + 10\cos\theta$ and evaluate $\int_0^{\pi/2} \cos^5\theta \, \mathrm{d}\theta$.

8. Let $\zeta = \operatorname{cis}(2\pi/5) = e^{2\pi i/5}$. Show that $1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4 = 0$. Show further that

$$(z - \zeta - \zeta^4)(z - \zeta^2 - \zeta^3) = z^2 + z - 1$$

and deduce that $\cos(2\pi/5) = (\sqrt{5} - 1)/4$.

9. Write down the seven roots of $z^7 + 1 = 0$. By considering the coefficient of z^6 in the factorization of $z^7 + 1$, show that

$$\cos\left(\frac{\pi}{7}\right) + \cos\left(\frac{3\pi}{7}\right) + \cos\left(\frac{5\pi}{7}\right) = \frac{1}{2}.$$

10.(i) Let $\zeta = \operatorname{cis}(\pi/7) = e^{i\pi/7}$. Simplify the expression $(\zeta - \zeta^6)(\zeta^3 - \zeta^4)(\zeta^5 - \zeta^2)$. Hence show that

$$\cos(\pi/7)\cos(3\pi/7)\cos(5\pi/7) = -1/8.$$

(ii) Given that $\cos 7\theta = 64\cos^7\theta - 112\cos^5\theta + 56\cos^3\theta - 7\cos\theta$, rederive the result of (i).

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Optional Extension Questions

11. (i) Let a, b, c be positive real numbers. By expanding $(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ and considering the second factor as a quadratic in a, show that

$$a^3 + b^3 + c^3 \geqslant 3abc \quad \text{if } a, b, c > 0.$$

(ii) Let $\omega^3 = 1$, $\omega \neq 1$. Show for any real numbers a, b, c that

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c).$$

Hence find all real numbers a, b, c which satisfy $a^3 + b^3 + c^3 = 3abc$.

12. (i) Let p, q be complex numbers. Show that *Viète's substitution* $z = w - p/(3w)$ turns the equation $z^3 + pz = q$ into the quadratic in w^3.
(ii) Use Viète's substitution to solve the cubic $z^3 - 12z + 8 = 0$.

13. (i) Let $P(z)$ be a polynomial with (possibly repeated) roots $\alpha_1, \alpha_2, \ldots, \alpha_k$. Show that

$$\frac{P'(z)}{P(z)} = \frac{1}{z-\alpha_1} + \frac{1}{z-\alpha_2} + \cdots + \frac{1}{z-\alpha_k}.$$

(ii) Deduce that if $\operatorname{Im} \alpha_i > 0$ for each i, then $\operatorname{Im} \beta > 0$ for any root β of $P'(z)$.
(iii) Deduce further that if all the roots α of a polynomial $P(z)$ satisfy $|\alpha| < R$ then all the roots β of $P'(z)$ satisfy $|\beta| < R$.

14. (i) Show, for any complex number z and positive integer n, that

$$z^{2n} - 1 = (z^2 - 1) \prod_{k=1}^{n-1} \left\{ z^2 - 2z \cos\left(\frac{k\pi}{n}\right) + 1 \right\}.$$

(ii) Deduce that for any real θ that

$$\sin n\theta = 2^{n-1} \sin\theta \prod_{k=1}^{n-1} \left\{ \cos\theta - \cos\left(\frac{k\pi}{n}\right) \right\}.$$

(iii) Determine

$$\prod_{k=1}^{2n} \cos\left(\frac{k\pi}{2n+1}\right).$$

15. Let $k > 0$.
(i) Sketch the curve C_k with equation $|z + 1/z| = k$.
(ii) What are the extreme values of $|z|$ on C_k?
(iii) Show that the curve C_2 consists of precisely two circles.